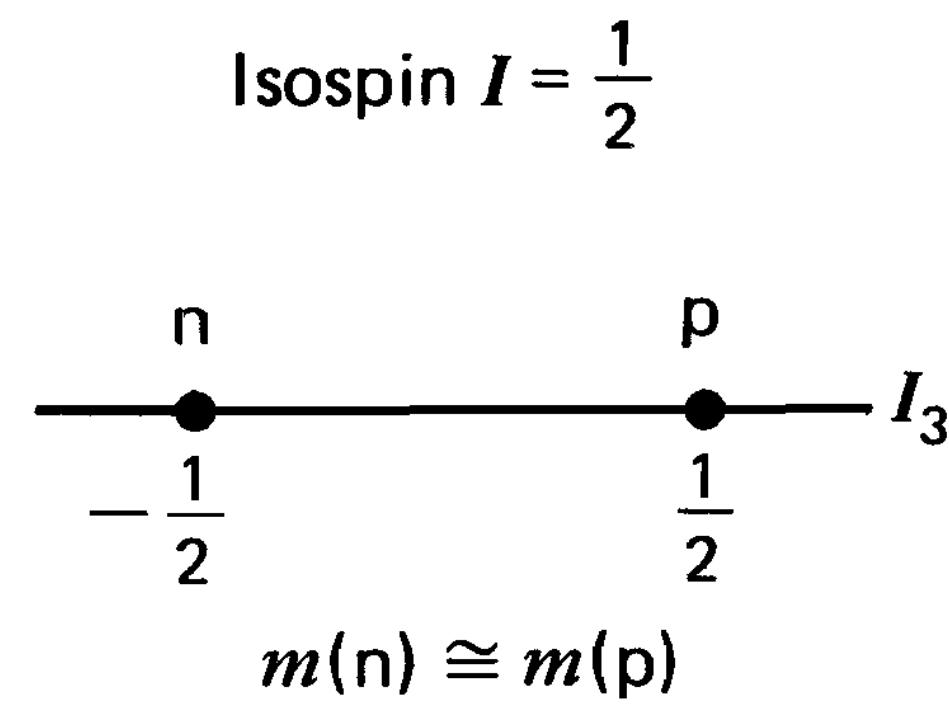
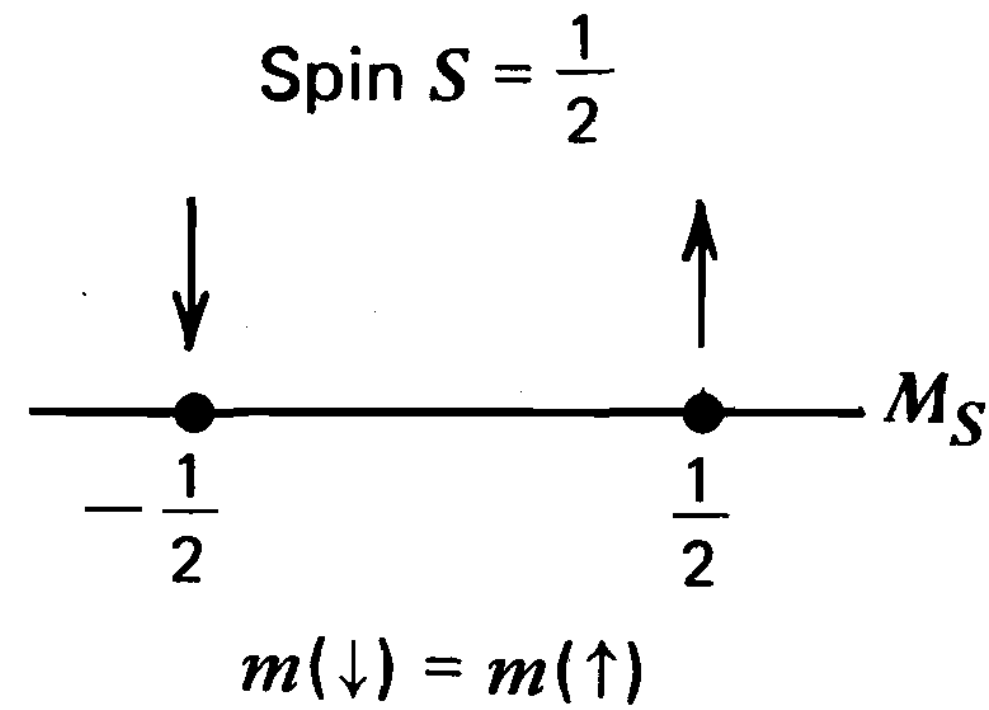


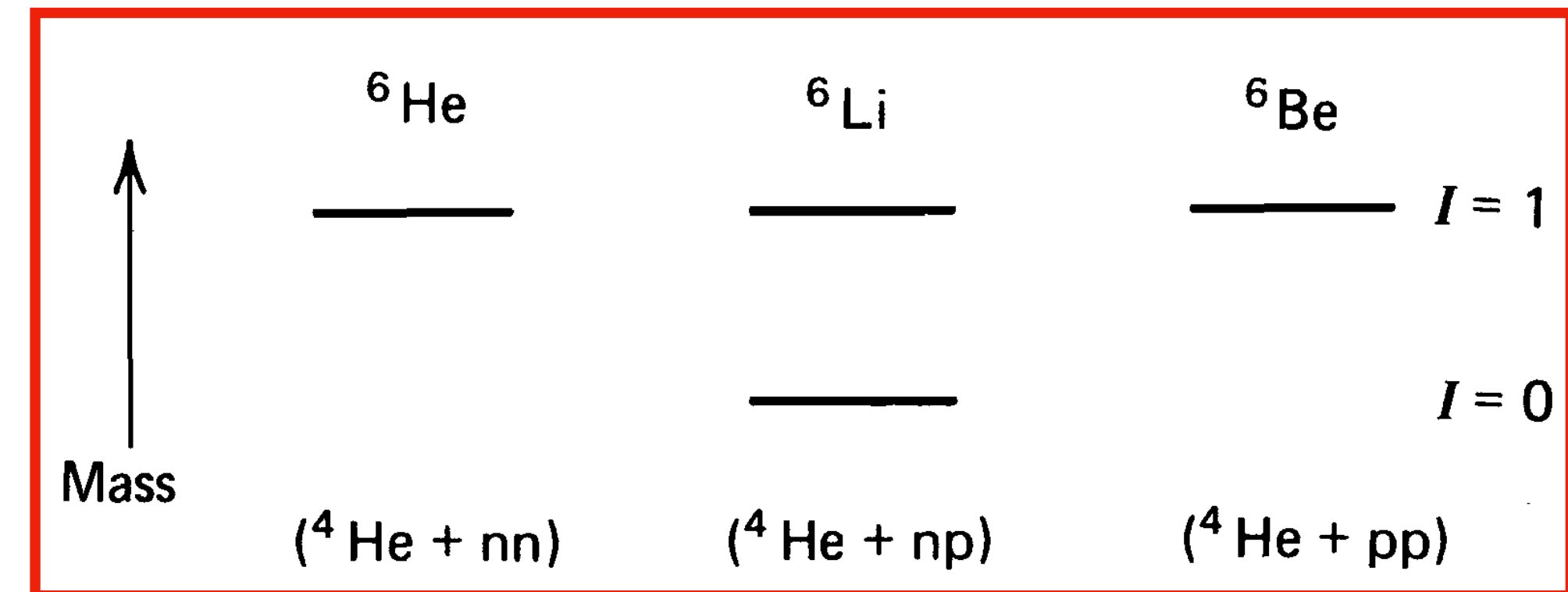
Introduction to Particle Physics

Isospin (likespin) Symmetry



Each nucleon is similarly postulated to have isospin $I = \frac{1}{2}$, with $I_3 = \pm \frac{1}{2}$ for protons and neutrons, respectively. $I = 1$ and $I = 0$ states of the nucleon–nucleon system can be constructed in exact analogy to spin:

$$\begin{cases} |I = 1, I_3 = 1\rangle = pp \\ |I = 1, I_3 = 0\rangle = \sqrt{\frac{1}{2}} (pn + np) \\ |I = 1, I_3 = -1\rangle = nn \\ |I = 0, I_3 = 0\rangle = \sqrt{\frac{1}{2}} (pn - np). \end{cases}$$



Recap

QM in 3D - Angular Momentum

The operators L_x and L_y do not commute;

$$\begin{aligned}[L_x, L_y] &= [yp_z - zp_y, zp_x - xp_z] \\ &= [yp_z, zp_x] - [yp_z, xp_z] - [zp_y, zp_x] + [zp_y, xp_z] \\ &= yp_x[p_z, z] + xp_y[z, p_z] = i\hbar(xp_y - yp_x) = i\hbar L_z.\end{aligned}$$

$$[L_x, L_y] = i\hbar L_z; \quad [L_y, L_z] = i\hbar L_x; \quad [L_z, L_x] = i\hbar L_y.$$

$$L_x, L_y \text{ and } L_z \text{ are incompatible operators} \quad \sigma_{L_x}^2 \sigma_{L_y}^2 \geq \left(\frac{1}{2i} \langle i\hbar L_z \rangle \right)^2 = \frac{\hbar^2}{4} \langle L_z \rangle^2,$$

The square of the total angular momentum does commute with each of them.

$$L^2 \equiv L_x^2 + L_y^2 + L_z^2,$$

$$\begin{aligned}[L^2, L_x] &= [L_x^2, L_x] + [L_y^2, L_x] + [L_z^2, L_x] \\ &= L_y[L_y, L_x] + [L_y, L_x]L_y + L_z[L_z, L_x] + [L_z, L_x]L_z \\ &= L_y(-i\hbar L_z) + (-i\hbar L_z)L_y + L_z(i\hbar L_y) + (i\hbar L_y)L_z \\ &= 0.\end{aligned}$$

$$[L^2, L_x] = 0, \quad [L^2, L_y] = 0, \quad [L^2, L_z] = 0, \text{ or, more compactly, } [L^2, \mathbf{L}] = 0.$$

Let's try to find simultaneous eigenstates of L^2 and L_z

$$L^2 f = \lambda f \quad \text{and} \quad L_z f = \mu f.$$

Let's use the same old trick!

$$L_{\pm} \equiv L_x \pm iL_y$$

$$\begin{aligned}[L_z, L_{\pm}] &= [L_z, L_x] \pm i[L_z, L_y] = i\hbar L_y \pm i(-i\hbar L_x) \\ &= \pm \hbar L_{\pm}.\end{aligned}$$

and

$$[L^2, L_{\pm}] = 0.$$

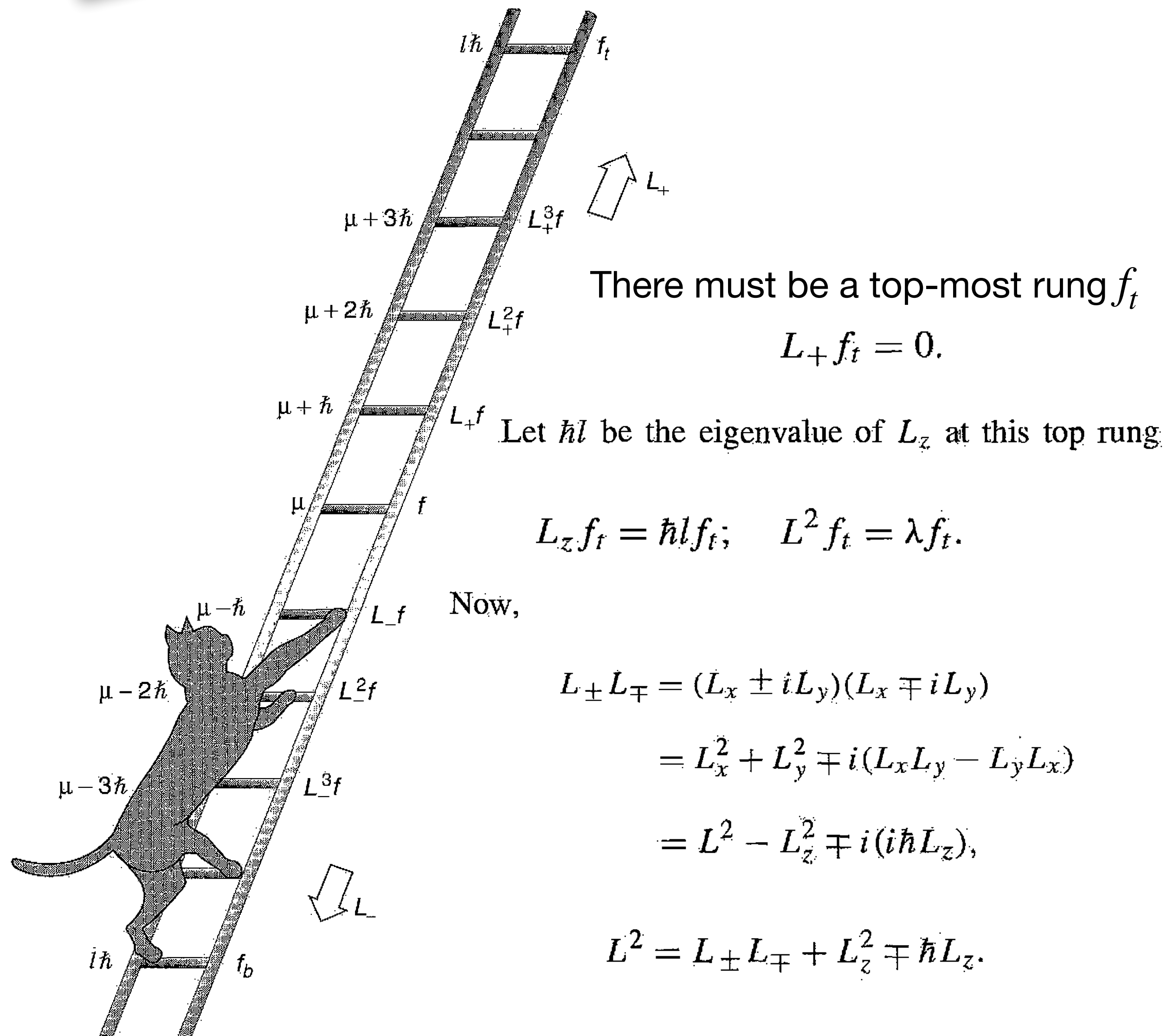
If f is an eigenfunction of L^2 and L_z then so is $L_{\pm}f$

$$L^2(L_{\pm}f) = L_{\pm}(L^2f) = L_{\pm}(\lambda f) = \lambda(L_{\pm}f),$$

$$\begin{aligned}L_z(L_{\pm}f) &= (L_z L_{\pm} - L_{\pm} L_z)f + L_{\pm} L_z f \\ &= \pm \hbar L_{\pm}f + L_{\pm}(\mu f) \\ &= (\mu \pm \hbar)(L_{\pm}f).\end{aligned}$$

Recap

QM in 3D - Angular Momentum



$$L^2 f_t = (L_- L_+ + L_z^2 + \hbar L_z) f_t = (0 + \hbar^2 l^2 + \hbar^2 l) f_t = \hbar^2 l(l+1) f_t,$$

or, $\lambda = \hbar^2 l(l+1).$

There must be a bottom-most rung f_b

Let $\hbar \bar{l}$ be the eigenvalue of L_z at this bottom rung:

$$L_z f_b = \hbar \bar{l} f_b; \quad L^2 f_b = \lambda f_b.$$

$$L^2 f_b = (L_+ L_- + L_z^2 - \hbar L_z) f_b = (0 + \hbar^2 \bar{l}^2 - \hbar^2 \bar{l}) f_b = \hbar^2 \bar{l}(\bar{l}-1) f_b,$$

or, $\lambda = \hbar^2 \bar{l}(\bar{l}-1).$

Hence, $l(l+1) = \bar{l}(\bar{l}-1),$

$\bar{l} = l+1$ (absurd) or, $\bar{l} = -l.$

Recap

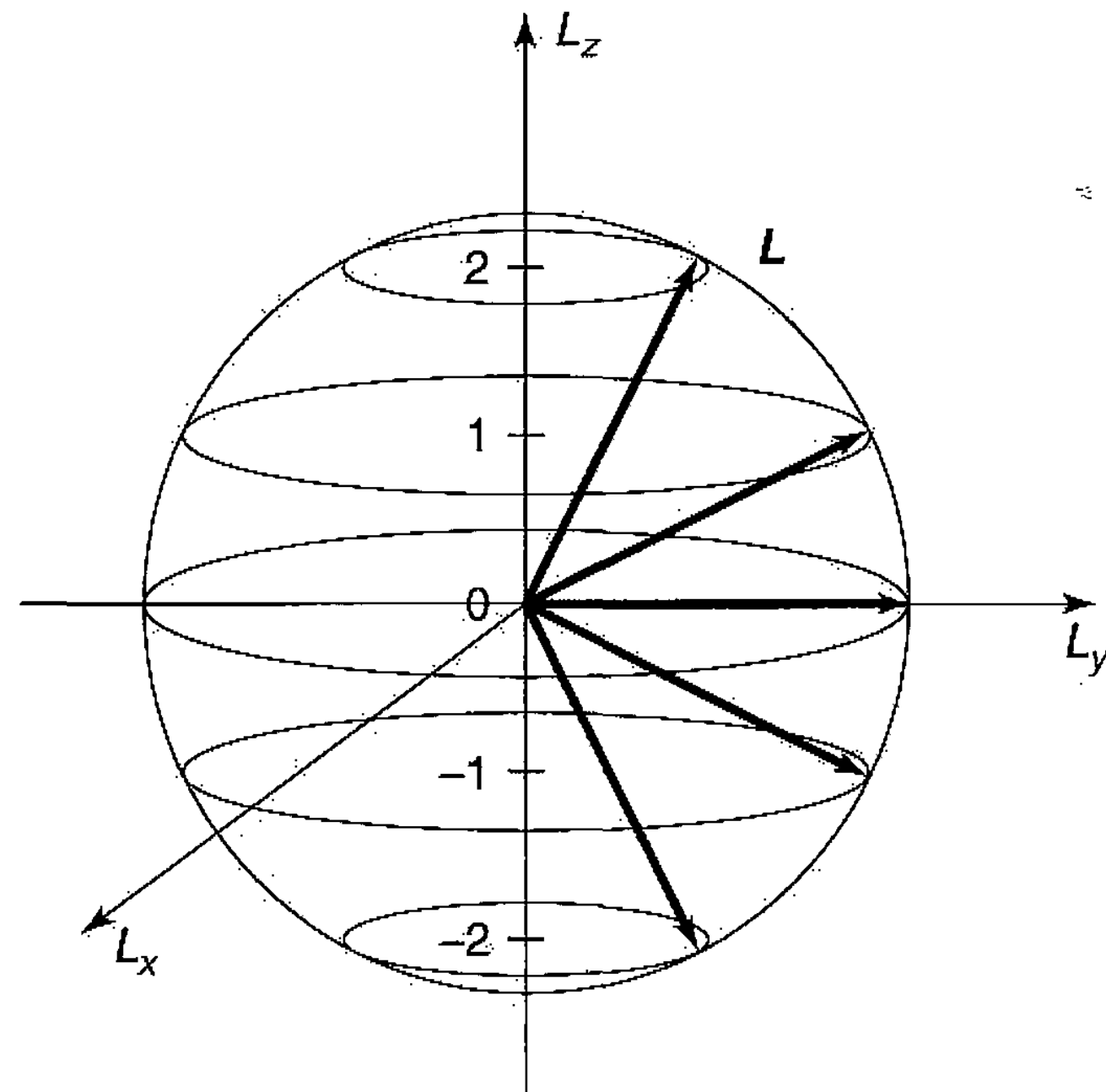
QM in 3D - Angular Momentum

The eigenvalues of L_z are $m\hbar$, where m goes from $-l$ to $+l$ in N integer steps, i.e., $l = N/2$ (an integer or a half-integer).

$$L^2 f_l^m = \hbar^2 l(l+1) f_l^m; \quad L_z f_l^m = \hbar m f_l^m,$$

$$l = 0, 1/2, 1, 3/2, \dots; \quad m = -l, -l+1, \dots, l-1, l.$$

For a given value of l , there are $2l+1$ different values of m (i.e., $2l+1$ “rungs” on the “ladder”).



Recap

QM in 3D - Spin

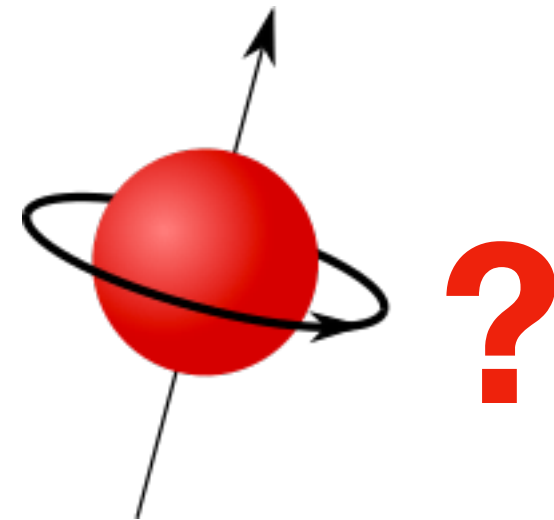
The *algebraic* theory of spin is a carbon copy of the theory of orbital angular momentum,

$$[S_x, S_y] = i\hbar S_z, \quad [S_y, S_z] = i\hbar S_x, \quad [S_z, S_x] = i\hbar S_y.$$

$$S^2 |s m\rangle = \hbar^2 s(s+1) |s m\rangle; \quad S_z |s m\rangle = \hbar m |s m\rangle;$$

$$S_{\pm} |s m\rangle = \hbar \sqrt{s(s+1) - m(m \pm 1)} |s (m \pm 1)\rangle,$$

$$s = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots; \quad m = -s, -s+1, \dots, s-1, s.$$



Spin-1/2

By far the most important case is $s = 1/2$, for this is the spin of the particles that make up ordinary matter (protons, neutrons, and electrons), as well as all quarks and all leptons. Moreover, once you understand spin 1/2, it is a simple matter to work out the formalism for any higher spin. There are just *two* eigenstates: $|\frac{1}{2} \frac{1}{2}\rangle$, which we call **spin up** (informally, $\uparrow$), and $|\frac{1}{2} (-\frac{1}{2})\rangle$, which we call **spin down** ($\downarrow$). Using these as basis vectors, the general state of a spin-1/2 particle can be expressed as a two-element column matrix (or **spinor**):

$$\chi = \begin{pmatrix} a \\ b \end{pmatrix} = a\chi_+ + b\chi_-,$$

with

$$\chi_+ = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

representing spin up, and

$$\chi_- = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

for spin down.

If we write $\mathbf{S}^2$ as a matrix with (as yet) undetermined elements,

$$\mathbf{S}^2 = \begin{pmatrix} c & d \\ e & f \end{pmatrix},$$

then the first equation says

$$\begin{pmatrix} c & d \\ e & f \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{3}{4}\hbar^2 \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \text{or} \quad \begin{pmatrix} c \\ e \end{pmatrix} = \begin{pmatrix} \frac{3}{4}\hbar^2 \\ 0 \end{pmatrix},$$

so $c = (3/4)\hbar^2$ and $e = 0$. The second equation says

$$\begin{pmatrix} c & d \\ e & f \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{3}{4}\hbar^2 \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \text{or} \quad \begin{pmatrix} d \\ f \end{pmatrix} = \begin{pmatrix} 0 \\ \frac{3}{4}\hbar^2 \end{pmatrix},$$

so $d = 0$ and $f = (3/4)\hbar^2$. *Conclusion:*

$$\mathbf{S}^2 = \frac{3}{4}\hbar^2 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

$$\mathbf{S}^2 \chi_+ = \frac{3}{4}\hbar^2 \chi_+ \quad \text{and} \quad \mathbf{S}^2 \chi_- = \frac{3}{4}\hbar^2 \chi_-.$$

Recap

QM in 3D - Spin 1/2

Similarly,

$$\mathbf{S}_z \chi_+ = \frac{\hbar}{2} \chi_+, \quad \mathbf{S}_z \chi_- = -\frac{\hbar}{2} \chi_-,$$

from which it follows that

$$\mathbf{S}_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Meanwhile,

$$\mathbf{S}_+ \chi_- = \hbar \chi_+, \quad \mathbf{S}_- \chi_+ = \hbar \chi_-, \quad \mathbf{S}_+ \chi_+ = \mathbf{S}_- \chi_- = 0,$$

so

$$\mathbf{S}_+ = \hbar \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{S}_- = \hbar \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Now $S_{\pm} = S_x \pm iS_y$, so $S_x = (1/2)(S_+ + S_-)$ and $S_y = (1/2i)(S_+ - S_-)$, and hence

$$\mathbf{S}_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \mathbf{S}_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}.$$

Since $\mathbf{S}_x$, $\mathbf{S}_y$, and $\mathbf{S}_z$ all carry a factor of $\hbar/2$, it is tidier to write $\mathbf{S} = (\hbar/2)\boldsymbol{\sigma}$, where

$$\sigma_x \equiv \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y \equiv \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z \equiv \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Pauli Spin Matrices

The eigenspinors of $\mathbf{S}_z$ are

$$\chi_+ = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \left(\text{eigenvalue} + \frac{\hbar}{2} \right); \quad \chi_- = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \left(\text{eigenvalue} - \frac{\hbar}{2} \right).$$

If you measure S_z on a particle in the general state χ you could get $+\hbar/2$, with probability $|a|^2$, or $-\hbar/2$, with probability $|b|^2$.

$$|a|^2 + |b|^2 = 1$$

Recap

QM in 3D

Addition of Angular Momenta

Suppose now that we have *two* spin-1/2 particles—for example, the electron and the proton in the ground state of hydrogen. Each can have spin up or spin down, so there are four possibilities in all:

$$\uparrow\uparrow, \uparrow\downarrow, \downarrow\uparrow, \downarrow\downarrow,$$

where the first arrow refers to the electron and the second to the proton. *Question:* What is the *total* angular momentum of the atom? Let

$$\mathbf{S} \equiv \mathbf{S}^{(1)} + \mathbf{S}^{(2)}.$$

Each of these four composite states is an eigenstate of S_z —the z components simply *add*:

$$\begin{aligned} S_z \chi_1 \chi_2 &= (S_z^{(1)} + S_z^{(2)}) \chi_1 \chi_2 = (S_z^{(1)} \chi_1) \chi_2 + \chi_1 (S_z^{(2)} \chi_2) \\ &= (\hbar m_1 \chi_1) \chi_2 + \chi_1 (\hbar m_2 \chi_2) = \hbar(m_1 + m_2) \chi_1 \chi_2, \end{aligned}$$

(note that $\mathbf{S}^{(1)}$ acts only on χ_1 , and $\mathbf{S}^{(2)}$ acts only on χ_2 ; this notation may not be elegant, but it does the job). So m (the quantum number for the composite system) is just $m_1 + m_2$:

$$\uparrow\uparrow: m = 1;$$

$$\uparrow\downarrow: m = 0;$$

$$\downarrow\uparrow: m = 0;$$

$$\downarrow\downarrow: m = -1.$$

At first glance, this doesn't look right: m is supposed to advance in integer steps, from $-s$ to $+s$, so it appears that $s = 1$ —but there is an “extra” state with $m = 0$. One way to untangle this problem is to apply the lowering operator, $S_- = S_-^{(1)} + S_-^{(2)}$ to the state $\uparrow\uparrow$,

$$\begin{aligned} S_-(\uparrow\uparrow) &= (S_-^{(1)} \uparrow) \uparrow + \uparrow (S_-^{(2)} \uparrow) \\ &= (\hbar \downarrow) \uparrow + \uparrow (\hbar \downarrow) = \hbar(\downarrow\uparrow + \uparrow\downarrow). \end{aligned}$$

Evidently the three states with $s = 1$ are (in the notation $|s m\rangle$):

$$\left\{ \begin{array}{l} |1 1\rangle = \uparrow\uparrow \\ |1 0\rangle = \frac{1}{\sqrt{2}}(\uparrow\downarrow + \downarrow\uparrow) \\ |1 -1\rangle = \downarrow\downarrow \end{array} \right\} \quad s = 1 \text{ (triplet)}.$$

Meanwhile, the orthogonal state with $m = 0$ carries $s = 0$:

$$\left\{ |0 0\rangle = \frac{1}{\sqrt{2}}(\uparrow\downarrow - \downarrow\uparrow) \right\} \quad s = 0 \text{ (singlet)}.$$

I claim, then, that the combination of two spin-1/2 particles can carry a total spin of 1 or 0, depending on whether they occupy the triplet or the singlet configuration.

Recap

QM in 3D

Addition of Angular Momenta

$$S^2 = (\mathbf{S}^{(1)} + \mathbf{S}^{(2)}) \cdot (\mathbf{S}^{(1)} + \mathbf{S}^{(2)}) = (S^{(1)})^2 + (S^{(2)})^2 + 2\mathbf{S}^{(1)} \cdot \mathbf{S}^{(2)}.$$

$$\begin{aligned} \mathbf{S}^{(1)} \cdot \mathbf{S}^{(2)}(\uparrow\downarrow) &= (S_x^{(1)} \uparrow)(S_x^{(2)} \downarrow) + (S_y^{(1)} \uparrow)(S_y^{(2)} \downarrow) + (S_z^{(1)} \uparrow)(S_z^{(2)} \downarrow) \\ &= \left(\frac{\hbar}{2} \downarrow\right) \left(\frac{\hbar}{2} \uparrow\right) + \left(\frac{i\hbar}{2} \downarrow\right) \left(\frac{-i\hbar}{2} \uparrow\right) + \left(\frac{\hbar}{2} \uparrow\right) \left(\frac{-\hbar}{2} \downarrow\right) \\ &= \frac{\hbar^2}{4}(2 \downarrow\uparrow - \uparrow\downarrow). \end{aligned}$$

Similarly,

$$\mathbf{S}^{(1)} \cdot \mathbf{S}^{(2)}(\downarrow\uparrow) = \frac{\hbar^2}{4}(2 \uparrow\downarrow - \downarrow\uparrow).$$

It follows that

$$\mathbf{S}^{(1)} \cdot \mathbf{S}^{(2)}|1\ 0\rangle = \frac{\hbar^2}{4} \frac{1}{\sqrt{2}}(2 \downarrow\uparrow - \uparrow\downarrow + 2 \uparrow\downarrow - \downarrow\uparrow) = \frac{\hbar^2}{4}|1\ 0\rangle,$$

and

$$\mathbf{S}^{(1)} \cdot \mathbf{S}^{(2)}|0\ 0\rangle = \frac{\hbar^2}{4} \frac{1}{\sqrt{2}}(2 \downarrow\uparrow - \uparrow\downarrow - 2 \uparrow\downarrow + \downarrow\uparrow) = -\frac{3\hbar^2}{4}|0\ 0\rangle.$$

we conclude that

$$S^2|1\ 0\rangle = \left(\frac{3\hbar^2}{4} + \frac{3\hbar^2}{4} + 2\frac{\hbar^2}{4}\right)|1\ 0\rangle = 2\hbar^2|1\ 0\rangle, \quad S^2|0\ 0\rangle = \left(\frac{3\hbar^2}{4} + \frac{3\hbar^2}{4} - 2\frac{3\hbar^2}{4}\right)|0\ 0\rangle = 0,$$

What we have just done (combining spin 1/2 with spin 1/2 to get spin 1 and spin 0) is the simplest example of a larger problem: If you combine spin s_1 with spin s_2 , what total spins s can you get? The answer is that you get every spin from $(s_1 + s_2)$ down to $(s_1 - s_2)$ —or $(s_2 - s_1)$, if $s_2 > s_1$ —in integer steps:

$$s = (s_1 + s_2), (s_1 + s_2 - 1), (s_1 + s_2 - 2), \dots, |s_1 - s_2|.$$

(Roughly speaking, the highest total spin occurs when the individual spins are aligned parallel to one another, and the lowest occurs when they are antiparallel.) For example, if you package together a particle of spin 3/2 with a particle of spin 2, you could get a total spin of 7/2, 5/2, 3/2, or 1/2, depending on the configuration. Another example: If a hydrogen atom is in the state ψ_{nlm} , the net angular momentum of the electron (spin plus orbital) is $l + 1/2$ or $l - 1/2$; if you now throw in spin of the *proton*, the atom's *total* angular momentum quantum number is $l + 1$, l , or $l - 1$ (and l can be achieved in two distinct ways, depending on whether the electron alone is in the $l + 1/2$ configuration or the $l - 1/2$ configuration).

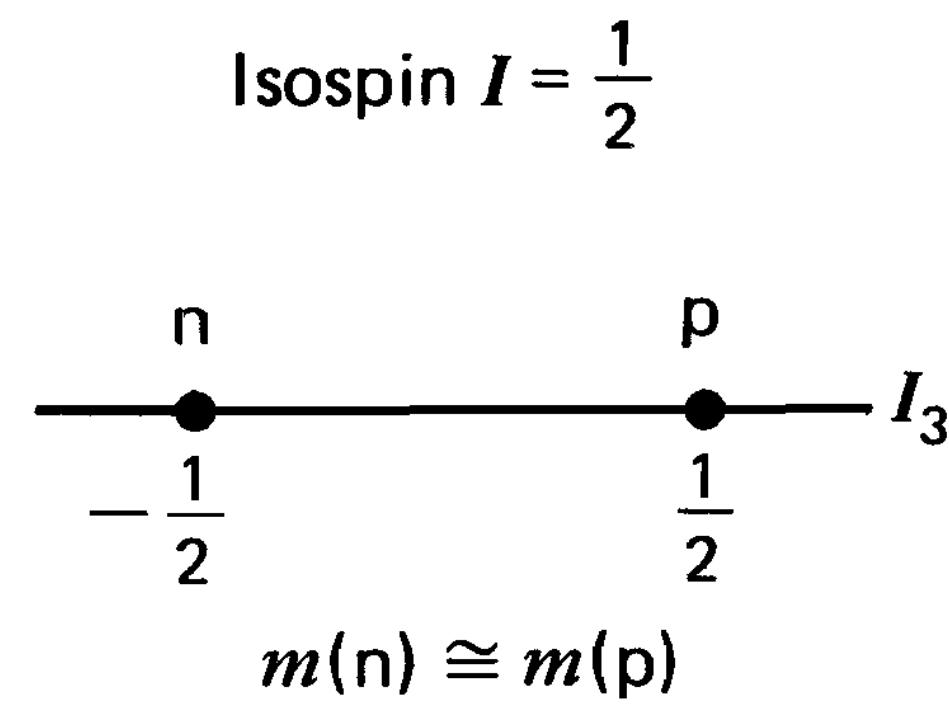
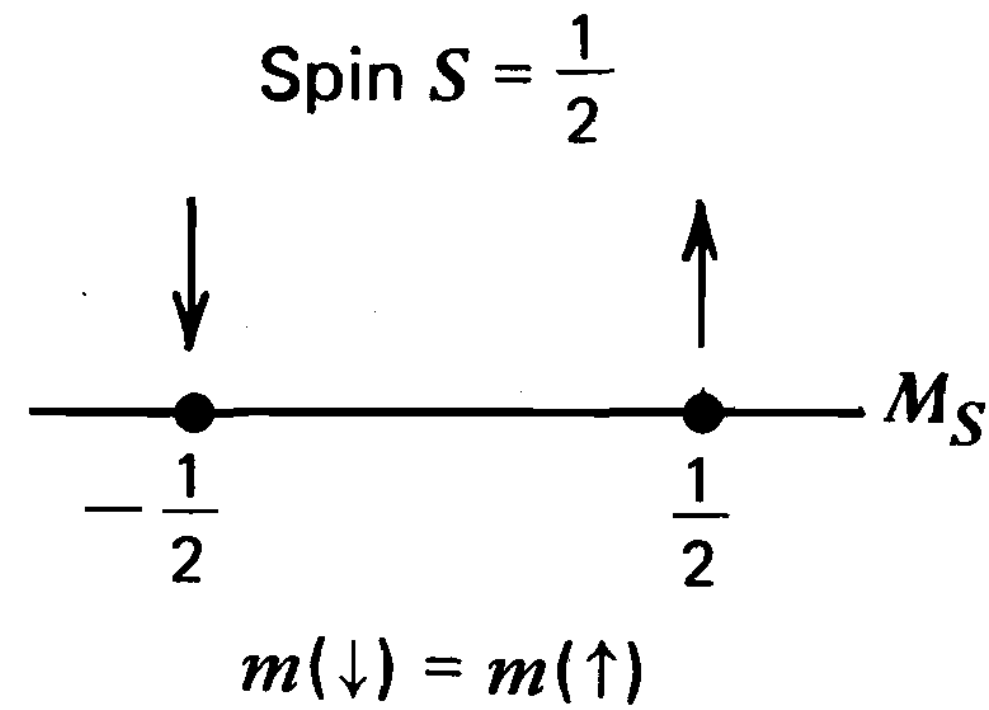
The combined state $|s\ m\rangle$ with total spin s and z -component m will be some linear combination of the composite states $|s_1\ m_1\rangle|s_2\ m_2\rangle$:

$$|s\ m\rangle = \sum_{m_1+m_2=m} C_{m_1 m_2 m}^{s_1 s_2 s} |s_1\ m_1\rangle |s_2\ m_2\rangle$$

(because the z components *add*, the only composite states that contribute are those for which $m_1 + m_2 = m$).

The constants $C_{m_1 m_2 m}^{s_1 s_2 s}$ are called **Clebsch-Gordan coefficients**.

Isospin (likespin) Symmetry



EXERCISE 2.3 Use isospin invariance to show that the reaction cross sections σ must satisfy

$$\frac{\sigma(\text{pp} \rightarrow \pi^+ \text{d})}{\sigma(\text{np} \rightarrow \pi^0 \text{d})} = 2,$$

given that the deuteron d has isospin $I = 0$ and the π has isospin $I = 1$.

Hint You may assume that the reaction rate is

$$\sigma \sim |\text{amplitude}|^2 \sim \sum_I |\langle I', I'_3 | A | I, I_3 \rangle|^2$$

where I and I' are the total isospin quantum numbers of the initial and final states, respectively, and $I = I'$ and $I_3 = I'_3$.

$$\begin{cases} |I = 1, I_3 = 1\rangle = \text{pp} \\ |I = 1, I_3 = 0\rangle = \sqrt{\frac{1}{2}} (\text{pn} + \text{np}) \\ |I = 1, I_3 = -1\rangle = \text{nn} \\ |I = 0, I_3 = 0\rangle = \sqrt{\frac{1}{2}} (\text{pn} - \text{np}). \end{cases}$$

Lie Group

The set of rotations of a system form a group, each rotation being an element of the group. Two successive rotations R_1 followed by R_2 (written as the “product” $R_2 R_1$) are equivalent to a single rotation (that is, to another group element). The set of rotations is closed under “multiplication.” There is an identity element (no rotation), and every rotation has an inverse (rotate back again). The “product” is not necessarily commutative, $R_1 R_2 \neq R_2 R_1$, but the associative law $R_3(R_2 R_1) = (R_3 R_2)R_1$ always holds. The rotation group is a continuous group in that each rotation can be labeled by a set of continuously varying parameters $(\alpha_1, \alpha_2, \alpha_3)$. These can be regarded as the components of a vector α directed along the axis of rotation with magnitude given by the angle of rotation.

The rotation group is a Lie group. The crucial property here is that every rotation can be expressed as the product of a succession of infinitesimal rotations (rotations arbitrarily close to the identity). The group is then completely defined by the “neighborhood of the identity.”

Lie Group

We do not want an experimental result to depend on the specific laboratory orientation of the system we are measuring. Rotations must therefore form a symmetry group of a system. They are a subset of the Lorentz transformations that can be performed on a system, namely, those transformations that leave it at rest. By definition, the physics is unchanged by a symmetry operation. In particular, these operations leave the transition probabilities of the system invariant. For example, suppose that under a rotation R the states of a system transform as

$$|\psi\rangle \rightarrow |\psi'\rangle = U|\psi\rangle.$$

The probability that a system described by $|\psi\rangle$ will be found in state $|\phi\rangle$ must be unchanged by R ,

$$|\langle\phi|\psi\rangle|^2 = |\langle\phi'|\psi'\rangle|^2 = |\langle\phi|U^\dagger U|\psi\rangle|^2,$$

and so U must be a unitary operator. The operators $U(R_1), U(R_2), \dots$, form a group with exactly the same structure as the original group $R_1, R_2, \dots$; they are said to form a unitary representation of the rotation group.

Moreover, the Hamiltonian is unchanged by a symmetry operation R of the system, and the matrix elements are preserved:

$$\langle\phi'|H|\psi'\rangle = \langle\phi|U^\dagger H U|\psi\rangle = \langle\phi|H|\psi\rangle,$$

so that

$$H = U^\dagger H U \quad \text{or} \quad [U, H] \equiv UH - HU = 0.$$

Lie Group

The transformation U has no explicit time dependence, and the equation of motion,

$$i\frac{d}{dt}|\psi(t)\rangle = H|\psi(t)\rangle,$$

is unchanged by the symmetry operation. As a consequence, the expectation value of U is a constant of the motion:

$$i\frac{d}{dt}\langle\psi(t)|U|\psi(t)\rangle = \langle\psi(t)|UH - HU|\psi(t)\rangle = 0.$$

All the group properties follow from considering infinitesimal rotations in the neighborhood of the identity. As an example, consider a rotation through an infinitesimal angle ϵ about the 3- (or z) axis. We may write, to first order in ϵ ,

$$U = 1 - i\epsilon J_3.$$

The operator J_3 is called the generator of rotations about the 3-axis. Now,

$$\begin{aligned} 1 &= U^\dagger U = (1 + i\epsilon J_3^\dagger)(1 - i\epsilon J_3) \\ &= 1 + i\epsilon(J_3^\dagger - J_3) + 0(\epsilon^2). \end{aligned}$$

Therefore, J_3 is hermitian and hence is a (quantum mechanical) observable.

Lie Group

To identify the observable J_3 , we consider the effect of a rotation on the wave function $\psi(\mathbf{r})$ describing the system. First, we must distinguish between two points of view. Either we may rotate the axes and keep the physical system fixed (the passive viewpoint) or we can keep the axes fixed and rotate the system (the active viewpoint). The viewpoints are equivalent; a rotation of the axes through an angle θ is the same as a rotation of the physical system by $-\theta$. We adopt the active viewpoint and rotate the physical system. The wave function ψ' describing the rotated state at $\mathbf{r}$ is then equal to the original function ψ at the point $R^{-1}\mathbf{r}$, which is transformed into $\mathbf{r}$ under the rotation R , that is,

$$\psi'(\mathbf{r}) = \psi(R^{-1}\mathbf{r}).$$

This specifies the one-to-one correspondence between ψ' and ψ , which we have written

$$\psi' = U\psi.$$

$$U\psi(x, y, z) = \psi(R^{-1}\mathbf{r}) \simeq \psi(x + \varepsilon y, y - \varepsilon x, z)$$

$$\simeq \psi(x, y, z) + \varepsilon \left(y \frac{\partial \psi}{\partial x} - x \frac{\partial \psi}{\partial y} \right)$$

$$= (1 - i\varepsilon(xp_y - yp_x))\psi.$$

$$[J_j, J_k] = i\varepsilon_{jkl}J_l$$

$$J^2 = J_1^2 + J_2^2 + J_3^2$$

$$[J^2, J_i] = 0 \quad \text{with } i = 1, 2, 3.$$

$$J^2|jm\rangle = j(j+1)|jm\rangle$$

$$J_3|jm\rangle = m|jm\rangle$$

$$U(\theta) = (U(\varepsilon))^n = \left(1 - i\frac{\theta}{n}J_3\right)^n \xrightarrow{n \rightarrow \infty} e^{-i\theta J_3}.$$

SU(2) Group

$$J_i = \frac{1}{2}\sigma_i \quad \text{with } i = 1, 2, 3,$$

where σ_i are the Pauli matrices

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The basis (or set of base states) for this representation is conventionally chosen to be the eigenvectors of σ_3 , that is, the column vectors

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

describing a spin- $\frac{1}{2}$ particle of spin projection up ($m = +\frac{1}{2}$ or $\uparrow$) and spin projection down ($m = -\frac{1}{2}$ or $\downarrow$) along the 3-axis, respectively.

The Pauli matrices σ_i are hermitian, and the transformation matrices

$$U(\theta_i) = e^{-i\theta_i\sigma_i/2}$$

are unitary. The set of all unitary 2×2 matrices is known as the group $U(2)$. However, $U(2)$ is larger than the group of matrices $U(\theta_i)$, since the generators σ_i all have zero trace. Now, for any hermitian traceless matrix σ , we can show that

$$\det(e^{i\sigma}) = e^{i\text{Tr}(\sigma)} = 1.$$

SU(2) Group

A state $|jm\rangle$ is transformed under a rotation through an angle θ about the 2-axis into a linear combination of the $2j + 1$ states $|jm'\rangle$, with $m' = -j, -j + 1, \dots, j$:

$$e^{-i\theta J_2}|jm\rangle = \sum_{m'} d_{m'm}^j(\theta)|jm'\rangle,$$

where the coefficients $d_{m'm}^j$ are written in conventional notation and are frequently called rotation matrices. we see the states having the same j but all possible m values transform among themselves under rotations. In fact, all the $2j + 1$ states are mixed by rotations. They form the basis of a $(2j + 1)$ -dimensional irreducible representation of the rotation group. The set of states is called a multiplet.

Since the unit determinant is preserved in matrix multiplication, the set of traceless unitary 2×2 matrices form a subgroup, $SU(2)$, of $U(2)$. $SU(2)$ denotes the special unitary group in two dimensions. The set of transformation matrices $U(\theta_i)$ therefore form an $SU(2)$ group. The $SU(2)$ algebra is just the algebra of the generators J_i , There are thus 1, 2, 3, 4, ... dimensional representations of $SU(2)$ corresponding to $j = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$, respectively. The two-dimensional representation is, of course, just the σ -matrices themselves. It is called the fundamental representation of $SU(2)$, the representation from which all other representations can be built, as we will now show.

$$d_{m'm}^j(\theta) = \langle jm'|e^{-i\theta J_2}|jm\rangle$$

$$j = \frac{1}{2} \begin{cases} d_{++} = d_{--} = \cos \frac{1}{2}\theta \\ d_{-+} = -d_{+-} = \sin \frac{1}{2}\theta \end{cases}$$

Combining Representations

A composite system formed from two systems having angular momentum j_A and j_B may be described in terms of the basis

$$|j_A j_B m_A m_B\rangle \equiv |j_A m_A\rangle |j_B m_B\rangle.$$

However, the combined operator

$$\mathbf{J} = \mathbf{J}_A + \mathbf{J}_B$$

also satisfies the Lie algebra and it is the eigenvalues $J(J+1)$, M of J^2 , J_3 which are the conserved quantum numbers. In fact, the “product” of the two irreducible representations of dimension $2j_A + 1$ and $2j_B + 1$ may be decomposed into the sum of irreducible representations of dimension $2J + 1$ with

$$J = |j_A - j_B|, |j_A - j_B| + 1, \dots, j_A + j_B,$$

with basis $|j_A j_B JM\rangle$, where

$$M = m_A + m_B.$$

The last equality follows directly from the third component may be expressed in terms of the other by

One basis

$$|j_A j_B JM\rangle = \sum_{m_A, m_B} C(m_A m_B; JM) |j_A j_B m_A m_B\rangle,$$

$$\begin{aligned} 2 \otimes 2 &= 3 \oplus 1, \\ (2 \otimes 2) \otimes 2 &= (3 \otimes 2) \oplus (1 \otimes 2) \\ &= 4 \oplus 2 \oplus 2. \end{aligned}$$

SU(2) of Isospin

$$[I_j, I_k] = i\epsilon_{jkl}I_l \quad \mathbf{p} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{n} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

In general, the most positively charged particle is chosen to have the maximum value of I_3 .

The construction of antiparticle isospin multiplets requires care. It is well illustrated by a simple example. Consider a particular isospin transformation of the nucleon doublet, a rotation through π about the 2-axis. We obtain

$$\begin{pmatrix} \mathbf{p}' \\ \mathbf{n}' \end{pmatrix} = e^{-i\pi(\tau_2/2)} \begin{pmatrix} \mathbf{p} \\ \mathbf{n} \end{pmatrix} = -i\tau_2 \begin{pmatrix} \mathbf{p} \\ \mathbf{n} \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \mathbf{p} \\ \mathbf{n} \end{pmatrix}.$$

We define antinucleon states using the particle–antiparticle conjugation operator C ,

$$C\mathbf{p} = \bar{\mathbf{p}}, \quad C\mathbf{n} = \bar{\mathbf{n}}.$$

Applying C

$$\begin{pmatrix} \bar{\mathbf{p}}' \\ \bar{\mathbf{n}}' \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \bar{\mathbf{p}} \\ \bar{\mathbf{n}} \end{pmatrix}.$$

However, we want the antiparticle doublet to transform in exactly the same way as the particle doublet, so that we can combine particle and antiparticle states using the same Clebsch–Gordan coefficients, and so on. We must therefore make two changes. First, we must reorder the doublet so that the most positively charged particle has $I_3 = +\frac{1}{2}$, and then we must introduce a minus sign to keep the matrix transformation identical

$$\begin{pmatrix} -\bar{\mathbf{n}}' \\ \bar{\mathbf{p}}' \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} -\bar{\mathbf{n}} \\ \bar{\mathbf{p}} \end{pmatrix}.$$

That is, the antiparticle doublet $(-\bar{\mathbf{n}}, \bar{\mathbf{p}})$ transforms exactly as the particle doublet $(\mathbf{p}, \mathbf{n})$. This is a special property of $SU(2)$; it is not possible, for example, to arrange an $SU(3)$ triplet of antiparticles so that it transforms as the particle triplet.

$$\begin{cases} |I = 1, I_3 = 1\rangle = -\mathbf{p}\bar{\mathbf{n}} \\ |I = 1, I_3 = 0\rangle = \sqrt{\frac{1}{2}} (\mathbf{p}\bar{\mathbf{p}} - \mathbf{n}\bar{\mathbf{n}}) \\ |I = 1, I_3 = -1\rangle = \mathbf{n}\bar{\mathbf{p}} \\ |I = 0, I_3 = 0\rangle = \sqrt{\frac{1}{2}} (\mathbf{p}\bar{\mathbf{p}} + \mathbf{n}\bar{\mathbf{n}}). \end{cases}$$

SU(3)

The set of unitary 3×3 matrices with $\det U = 1$ form the group $SU(3)$. The generators may be taken to be any $3^2 - 1 = 8$ linearly independent traceless hermitian 3×3 matrices. Since it is possible to have only two of these traceless matrices diagonal, this is the maximum number of mutually commuting generators. This number is called the rank of the group, so that $SU(3)$ has rank 2 and $SU(2)$ has rank 1. It can be shown that the number of Casimir operators is equal to the rank of the group.

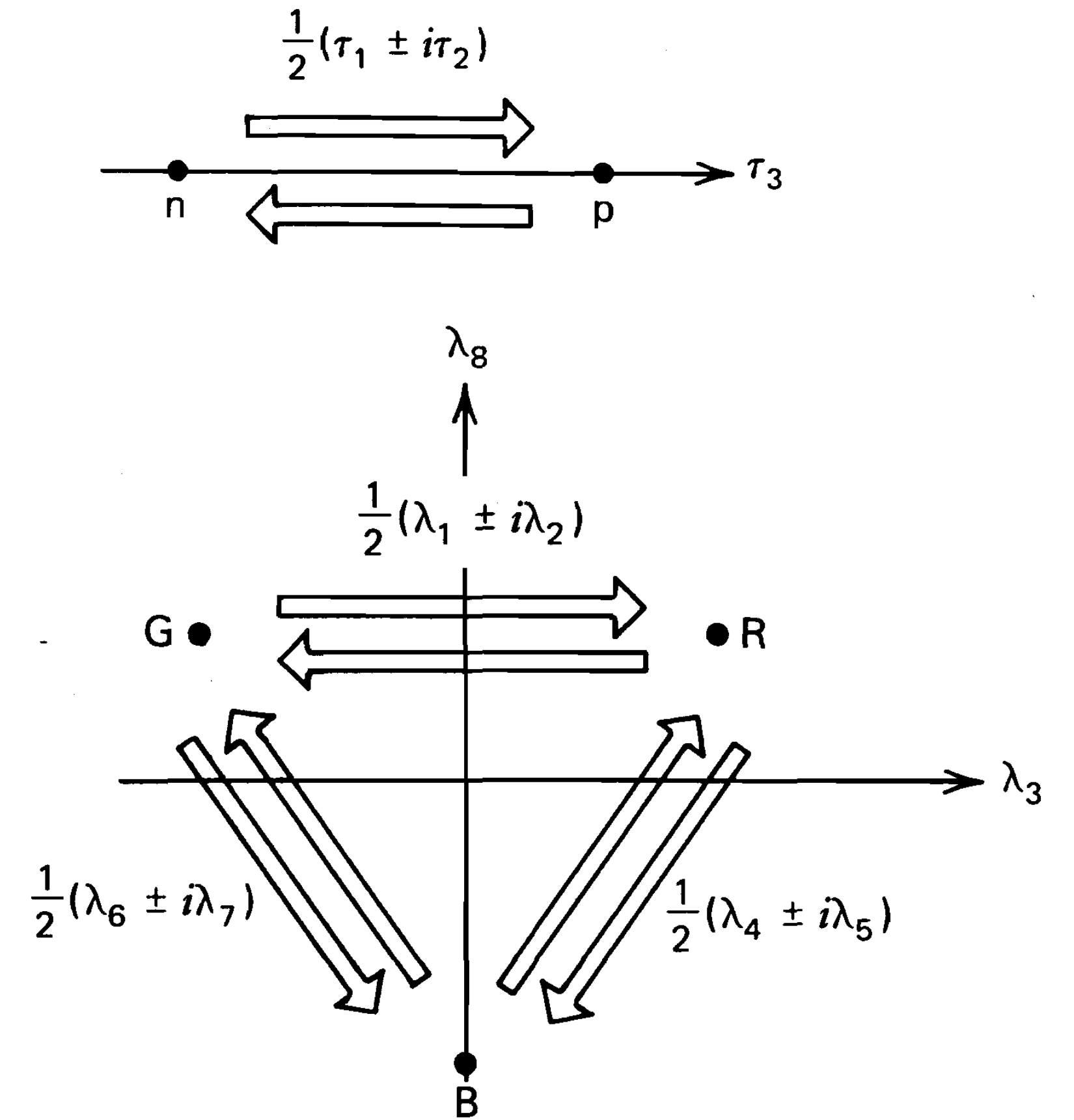
The fundamental representation of $SU(3)$ is a triplet. The three color charges of a quark, R, G, and B of Section 1.2, form the fundamental representation of an $SU(3)$ symmetry group. In this representation, the generators are 3×3 matrices. They are traditionally denoted λ_i , with $i = 1, \dots, 8$, and the diagonal matrices are taken to be

$$\lambda_3 = \begin{pmatrix} 1 & & \\ & -1 & \\ & & 0 \end{pmatrix}, \quad \lambda_8 = \sqrt{\frac{1}{3}} \begin{pmatrix} 1 & & \\ & 1 & \\ & & -2 \end{pmatrix} \quad (2.43)$$

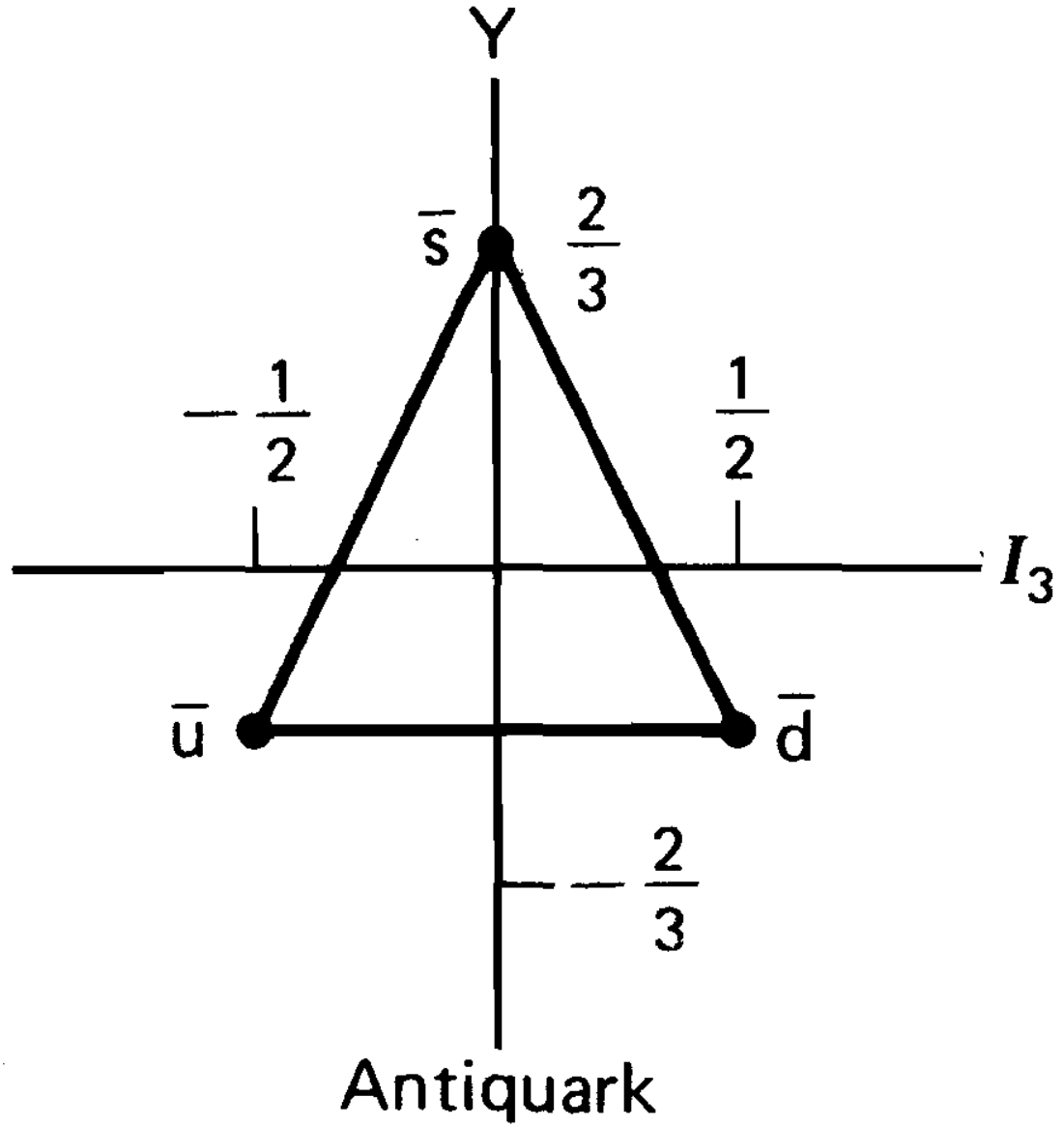
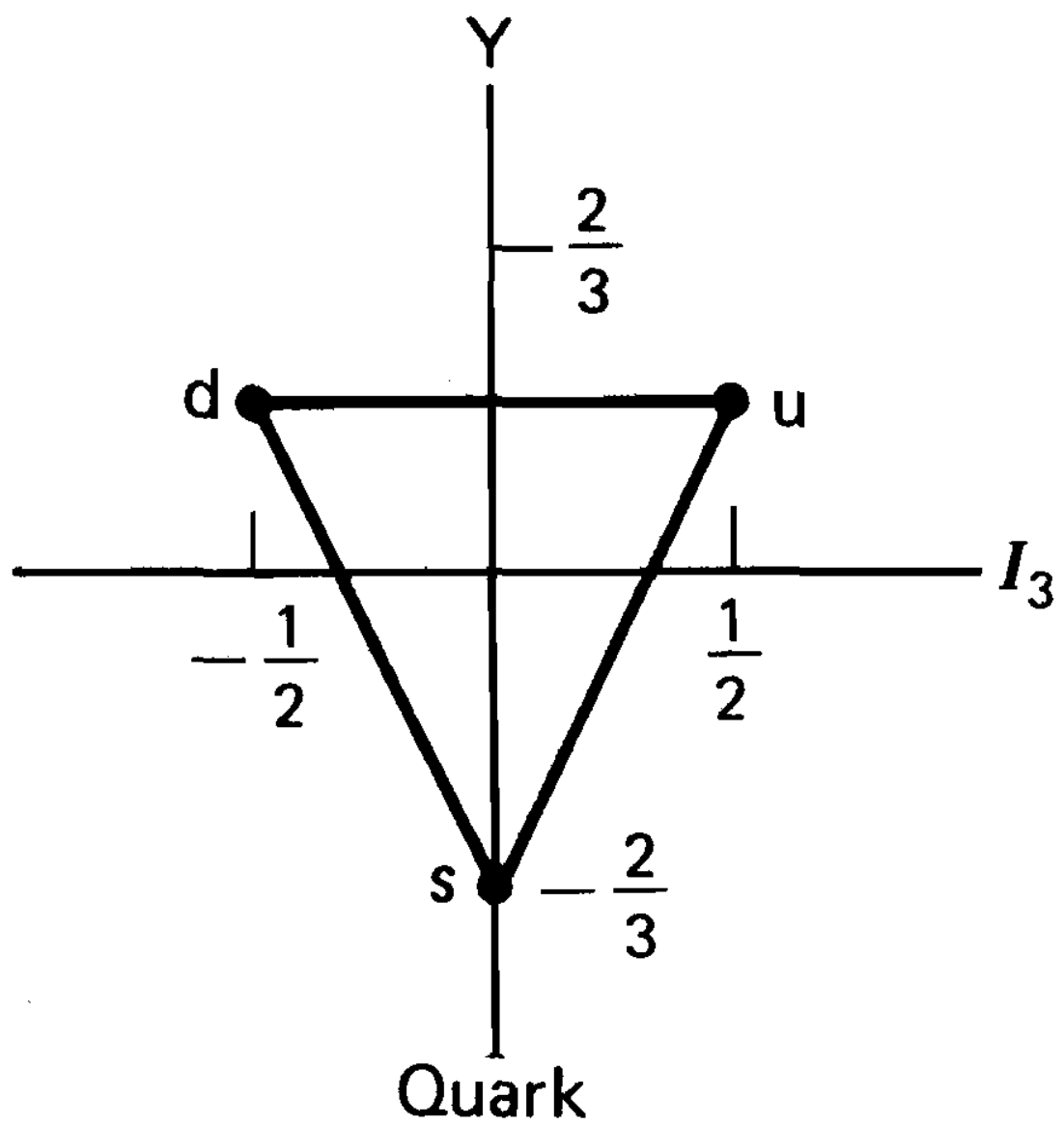
with simultaneous eigenvectors

$$\mathbf{R} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{G} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

$$\left[\frac{\lambda_i}{2}, \frac{\lambda_j}{2} \right] = i \sum_k f_{ijk} \frac{\lambda_k}{2} \quad \begin{aligned} f_{123} &= 1, & f_{458} &= f_{678} = \sqrt{3}/2, \\ f_{147} &= f_{165} = f_{246} = f_{257} = f_{345} = f_{376} &= \frac{1}{2}. \end{aligned}$$



SU(3) of Flavor: Isospin & Strangeness



$$Y \equiv B + S,$$

$$Q = I_3 + \frac{Y}{2}.$$

$$\begin{cases} |I = 1, I_3 = 1\rangle = -u\bar{d} \\ |I = 1, I_3 = 0\rangle = \sqrt{\frac{1}{2}}(u\bar{u} - d\bar{d}) \\ |I = 1, I_3 = -1\rangle = d\bar{u}, \end{cases}$$

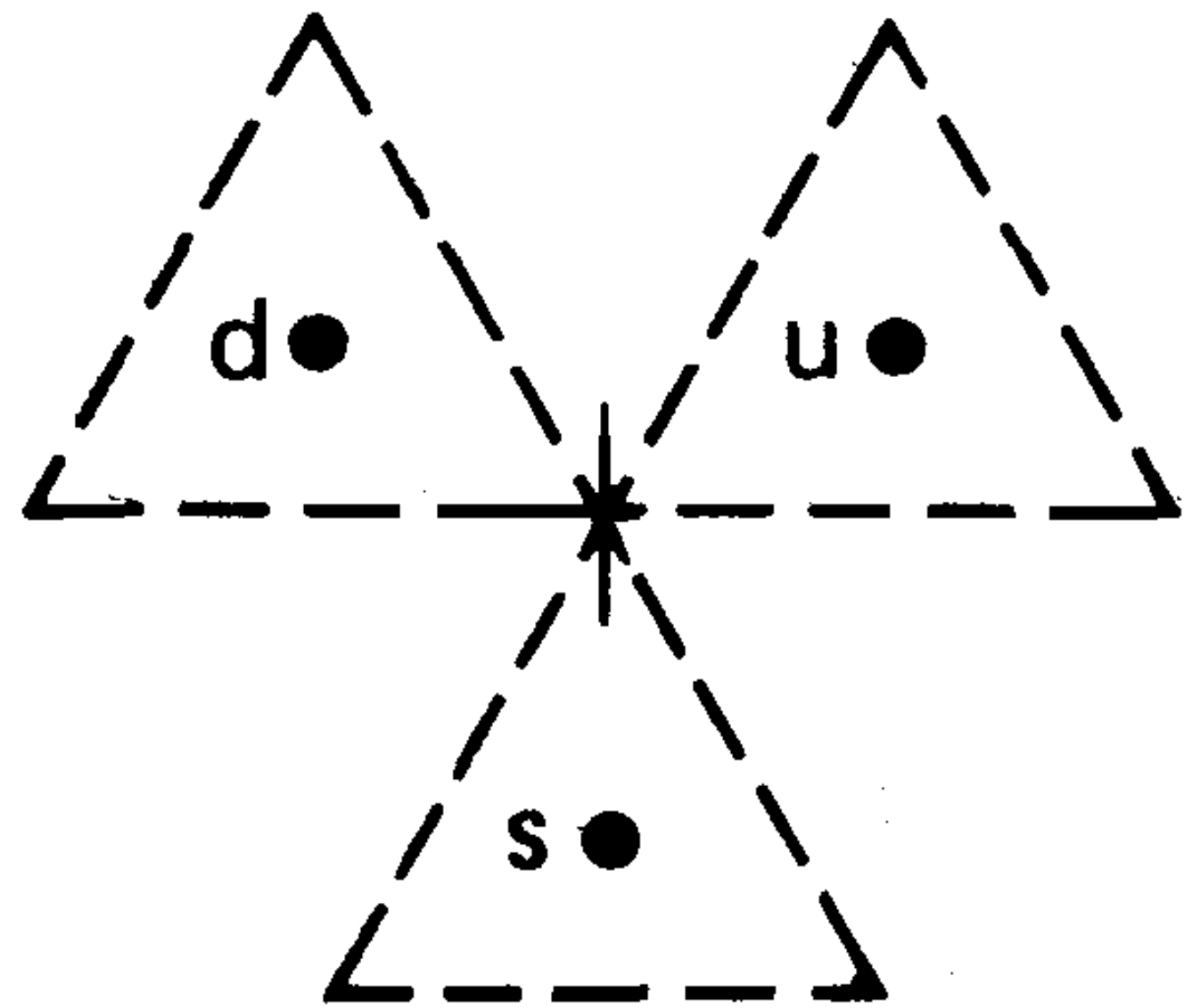
$$|I = 0, I_3 = 0\rangle = \sqrt{\frac{1}{2}}(u\bar{u} + d\bar{d}).$$

Quantum Numbers of the Quarks ($Y = B + S, Q = I_3 + Y/2$)

Quark	Spin	B	Q	I ₃	S	Y
u	1/2	1/3	2/3	1/2	0	1/3
d	1/2	1/3	-1/3	-1/2	0	1/3
s	1/2	1/3	-1/3	0	-1	-2/3

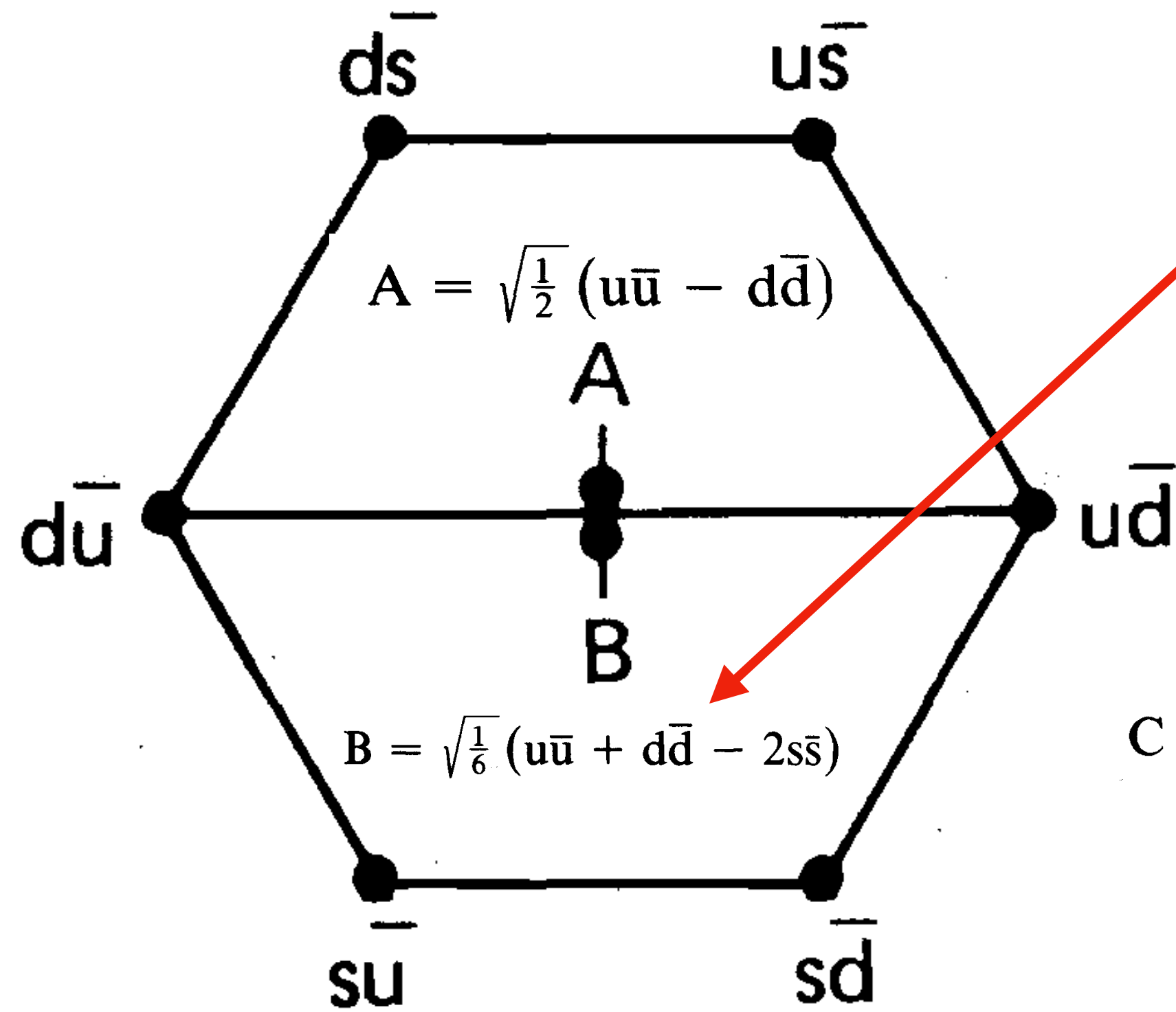
SU(3) of Flavor: Isospin & Strangeness

Meson Nonet



$$3 \otimes \bar{3}$$

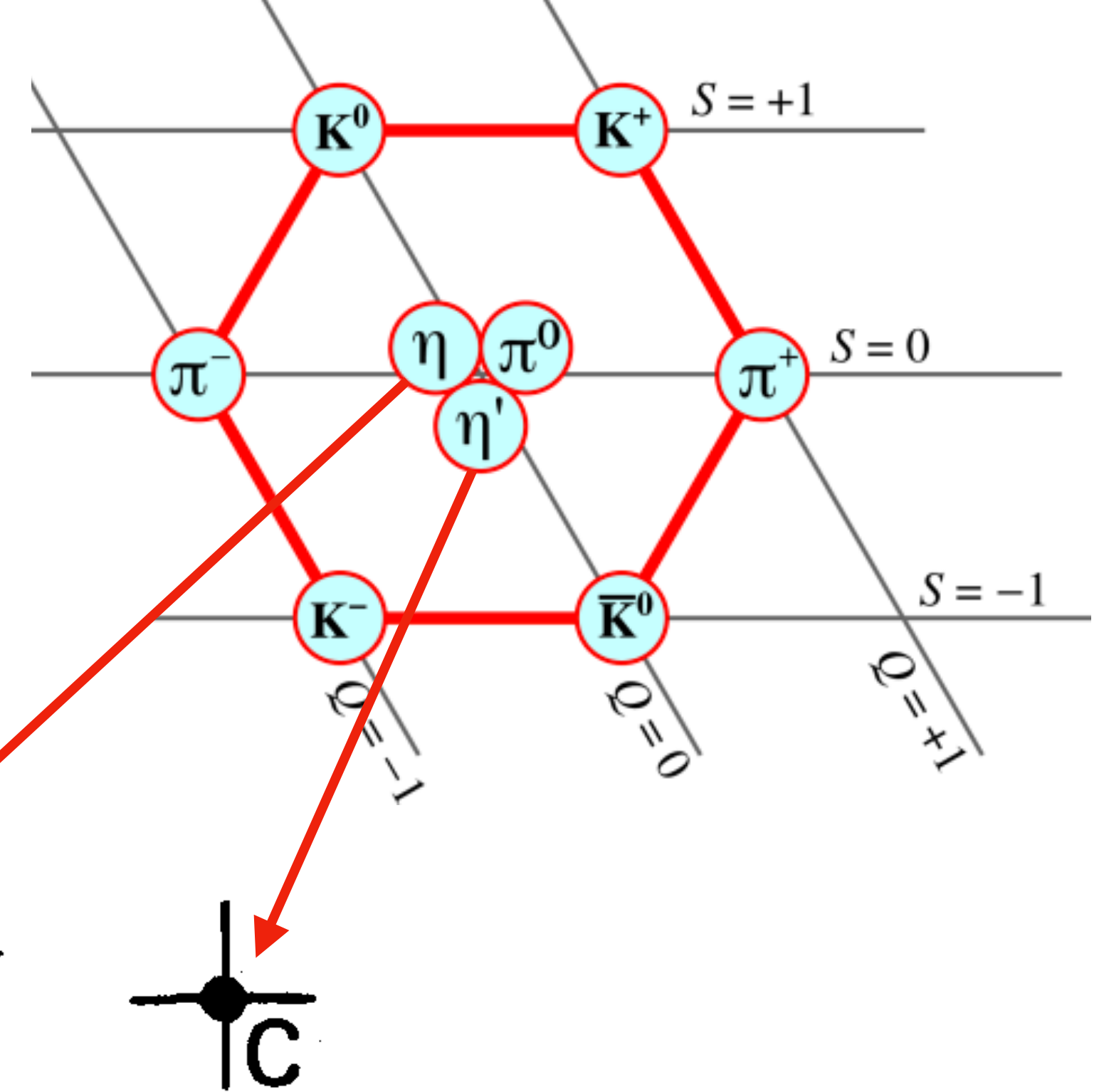
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$$C = \sqrt{\frac{1}{3}} (u\bar{u} + d\bar{d} + s\bar{s})$$

SU(3) of Flavor: Isospin & Strangeness

Baryon Multiplates

